

PD Module 3— Lab Manual: Floorplan

- 1 Invoke the tool
- 2 Open the library and block created in the previous class
- 3 Perform sanity checks
- 4 Create a floorplan, given
 - a Core Utilization
 - b Aspect Ratio
 - c Shape (Rectangular / Rectilinear)
 - d Space between core and die area
- 5 Placing the I/O ports
- 6 Placing Macros into the core area
- 7 Set Keepout margin and fix the macros
- 8 Add End Cap Cell
- 9 Add Tap cells
- 10 Check Legality

Task

Open the directory which was created in the previous lab

```
Linux > cd chiptop2.0
Linux > cd pd/work
```

Invoke the icc2 tool

```
Linux > icc2_shell -o ../logs/floorplan.log
```

Open the Library

```
icc2_shell > open_lib ../lib/chiptop
```

To see all the blocks present in the library

```
icc2_shell > list_blocks
```

Now, open the import done block to which netlist and SDC files were read in the previous lab

```
icc2_shell > open_block import_done
```

Open one more terminal in the scripts directory open, "floorplan.tcl" using "vi" or "gvim" editor command from the scripts directory and go through the file contents.

```
linux > gvim floorplan.tcl
```

Sanity Checks:

Check the design for netlist connectivity errors

```
icc2_shell > check_netlist
```

Lets check the timing constraints in the current design

```
icc2_shell > check_timing
```

Check if mismatches are present between the current design and reference libraries

```
icc2_shell > report_design_mismatch -verbose
```

View the information about the reference library

```
icc2_shell > report_lib [get_libs saed14hvt_ss0p6v125c]
```

Create Floorplan:

To open the Graphical User Interface (GUI)

```
icc2_shell > start_gui
```

Block Window (layout window) will open as shown in Fig 1.

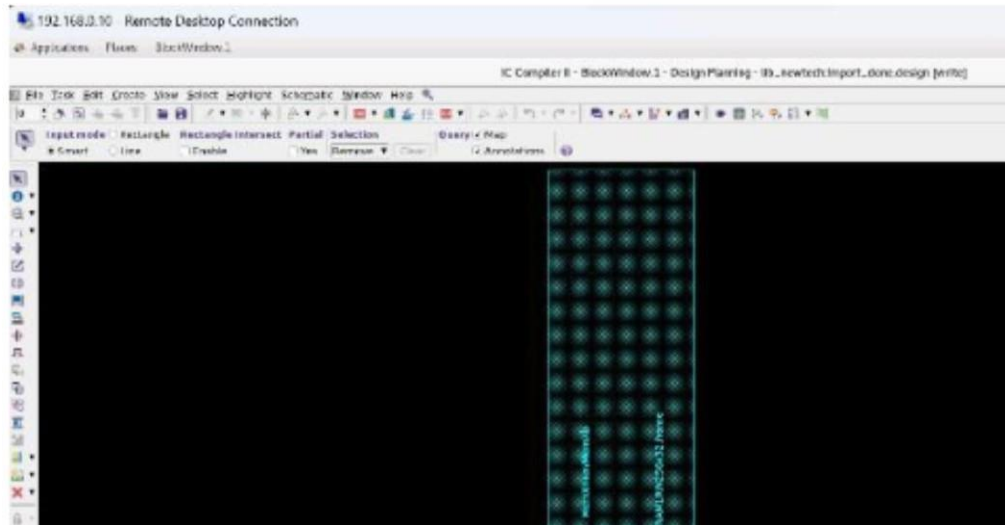


Fig 1.

The large rectangular shapes as shown in Fig 1 are the macros, this design has 4 such macros. In the toolbar, Click on Task → Design planning, a window will pop out. Select Floorplan Initialization as shown in Fig 2.

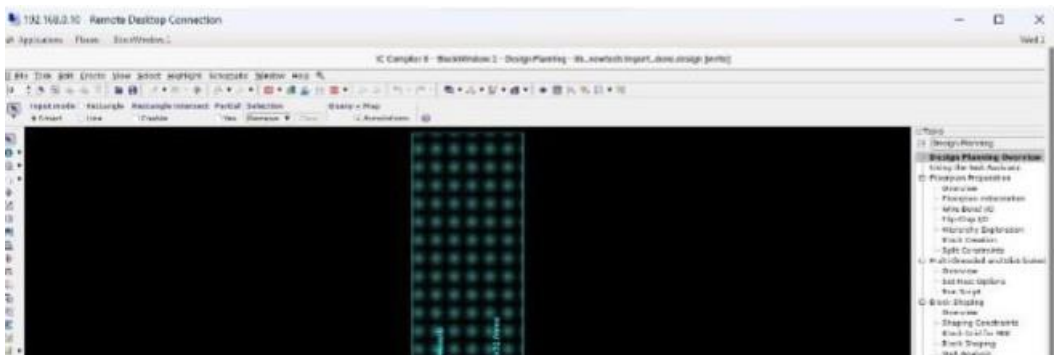


Fig 2.

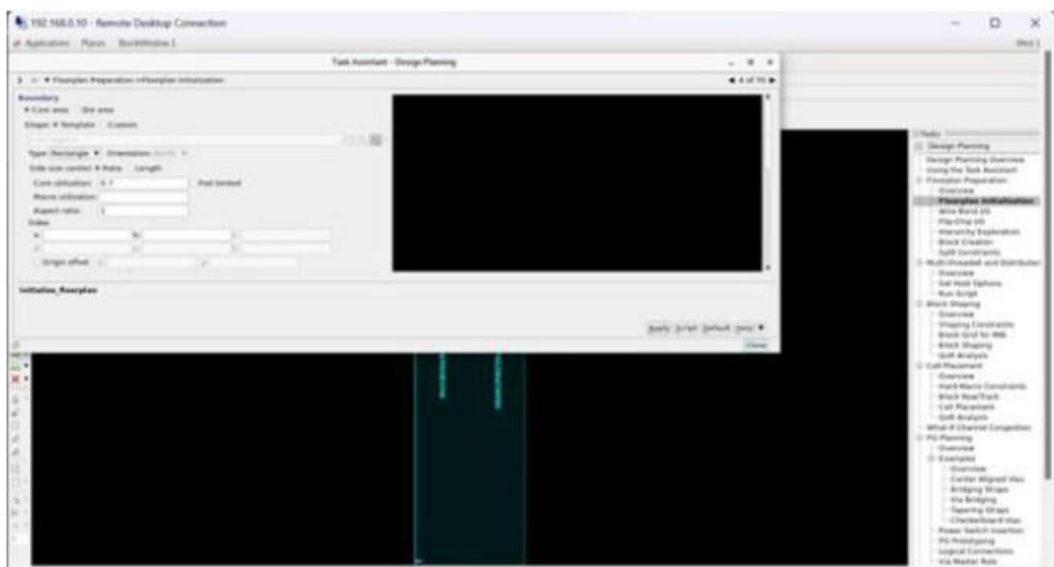


Fig 3 : Floorplan initialization

In Floorplan initialization window,

Select : Core Area

Shape : Template

Type : Rectangle

Side Size Control : Ratio

Core Utilization = 0.65

Aspect Ratio = 1

Sides a=1 b=0.7

Select : Die Boundary follows core shape

Uniform : 5

Click on Apply

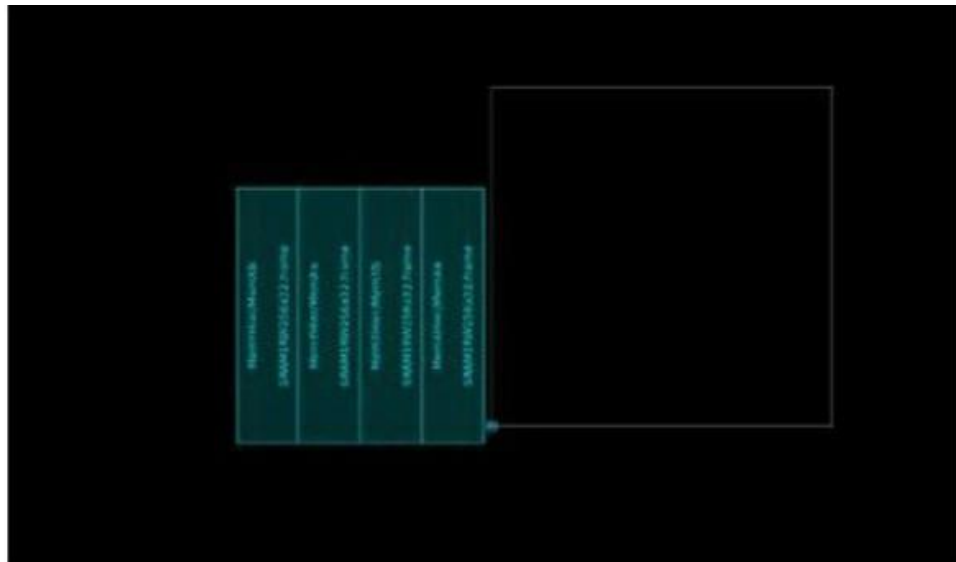


Fig 4 : After floorplan initialization

Placement of I/O ports:

Set the constraints for output pin placement.

```
icc2_shell> set_block_pin_constraints -self  
-allowed_layers {M3 M4}-sides 2
```

Perform output pin placement.

```
icc2_shell> place_pins -ports [get_ports -filter  
direction==out]
```

Set the constraints for input pin placement.

```
icc2_shell> set_block_pin_constraints -self  
-allowed_layers {M3 M4}-sides 4
```

Perform input pin placement.

```
icc2_shell> place_pins -ports [get_ports -filter  
direction==in]
```

Check whether the ports are fixed or not.

```
icc2_shell> get_attribute [get_ports *] is_fixed
```

Set all the ports to the fixed state if they are not.

```
icc2_shell> set_attribute [get_ports *] physical_status  
fixed
```

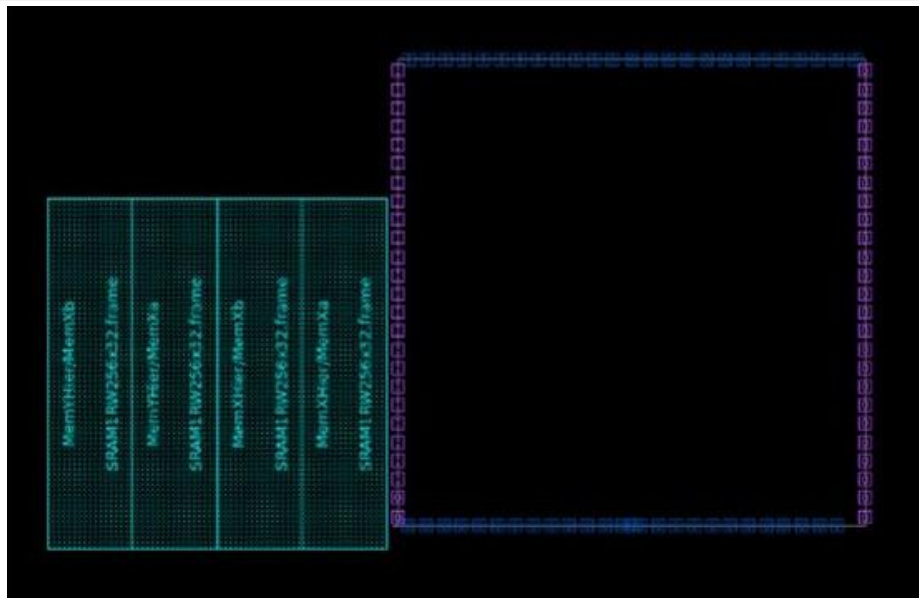


Fig 5 : Input/Output Ports Placement

Activity 3:

Look into all input / output ports of this project.

```
icc2_shell > get_ports
```

Compare with the ports present in the top module of the netlist.

Get the count of ports.

```
icc2_shell > sizeof_collection [get_ports *]
```

Placement of Macros:

Move the macros (select macro and press m to move) into core area, based on interconnections b/w the macros to macros and I/O ports (by enabling flylines), signal pins of macros facing towards the core. Use align and distribute options of the tool while placing macros.

Refer to Fig 6 to enable fly lines (Net Connections)

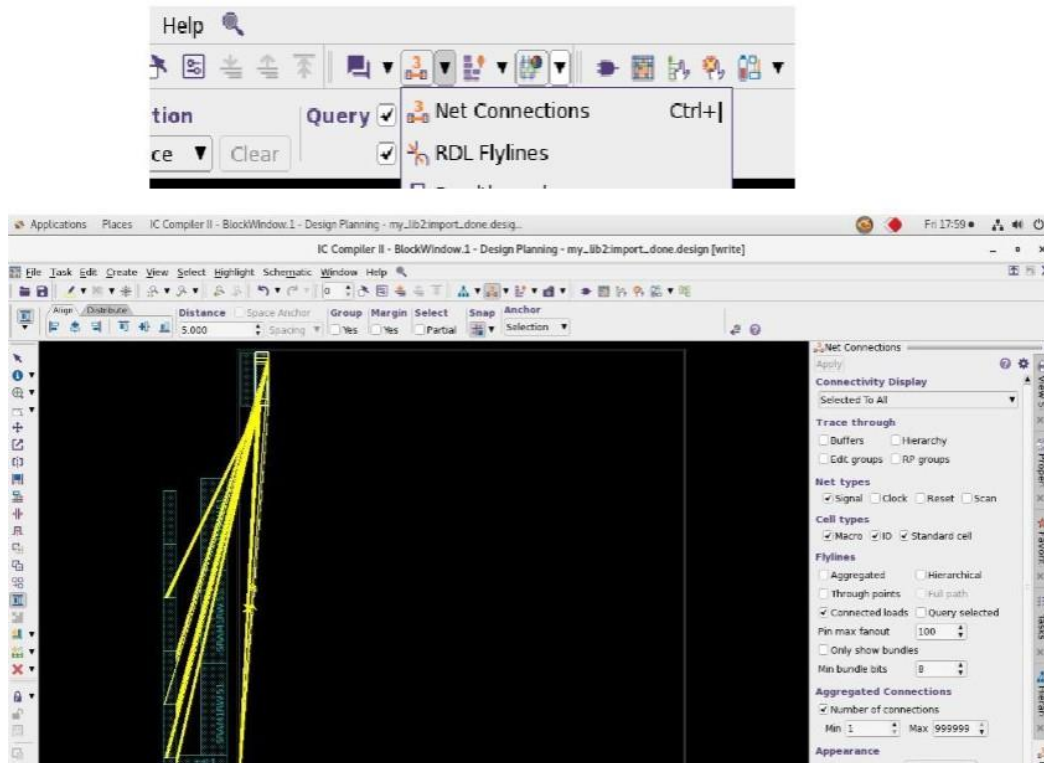


Fig 6.

Place all Macros in the core area (as shown in Fig 8)

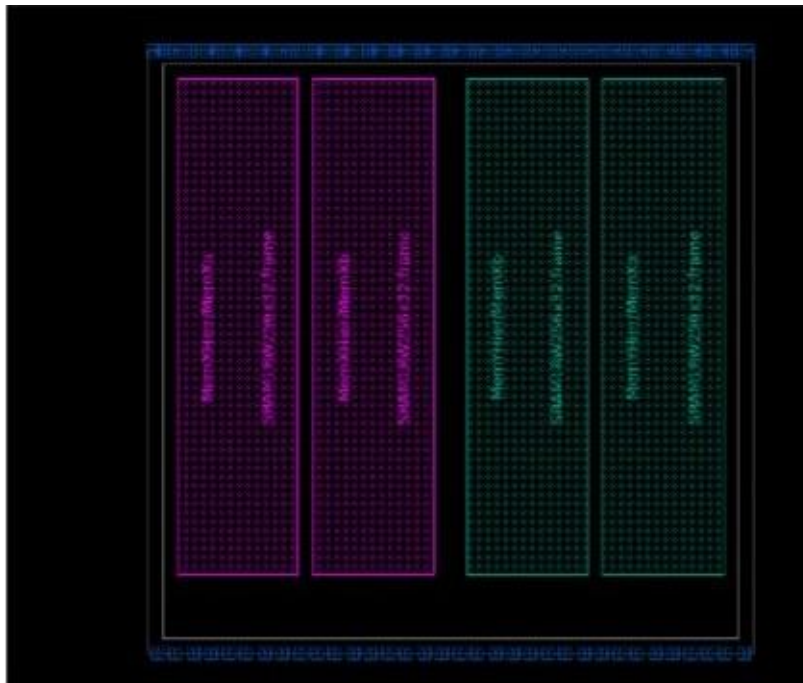


Fig 8 : After placement of Macros

Activity 4 :

List the Guidelines followed for placement of macros.

- 1.
- 2.
- 3.

We need to set the keepout margin across the macros so that standard cells are not placed close to the edge of Macros. Following command creates keepout margins for all macros with type hard.

```
icc2_shell> create_keepout_margin -type hard -outer {1.48
1.2 1.48 1.2} [get_cells -physical_context -filter
design_type==macro]
```

Activity 5:

What is keep out margin? Differentiate between Hard and Soft margins.

To fix the Macros:

Check whether Macros are fixed or not.

```
icc2_shell> get_attribute [get_cells -physical_context
-filter design_type==macro] is_fixed
```

Let's fix the macros if they are unfixed.

```
icc2_shell> set_attribute [get_cells -physical_context  
-filter design_type==macro] physical_status fixed
```

Activity 6:

Run the following command to unfix the fixed macros

```
icc2_shell> set_fixed_objects [get_cells -physical_context  
-filter design_type==macro] -unfix
```

Run the following command to check whether macros are placed or not.

```
icc2_shell> get_attribute [get_cells -physical_context  
-filter design_type==macro] physical_status
```

Add End Cap Cells:

End-cap cells are physical only cells which are added to identify end of rows in digital chips or blocks. They are also added to isolate any analog IPs and digital part of any chips.

First set the rules for placing the cells, like which boundary and the library cell to be used as End cap cell.

```
icc2_shell > set_boundary_cell_rules -top_boundary_cells  
saed14s*vt_ss0p72v125c/SAED*VT14_CAPT2  
-bottom_boundary_cells saed14vt_*VT14_CAPBIN13  
-left_boundary_cell saed14*vt_tt0p6v125c/  
SAED*VT14_CAPBTAP6 -prefix endcap
```

Now, compile to physically place the cells,

```
icc2_shell > compile_targeted_boundary_cells  
-target_objects [get_voltage_areas]
```

Add Tap Cells:

Tap cells are used to limit resistance between power or ground connections to wells of the substrate.

Create the tap cells:

```
icc2_shell > create_tap_cells -lib_cell saed14slvt_ss0p72v  
125c/SAEDSLVT14_CAPTTAPP6 -distance 30 -skip_fixed_cells
```

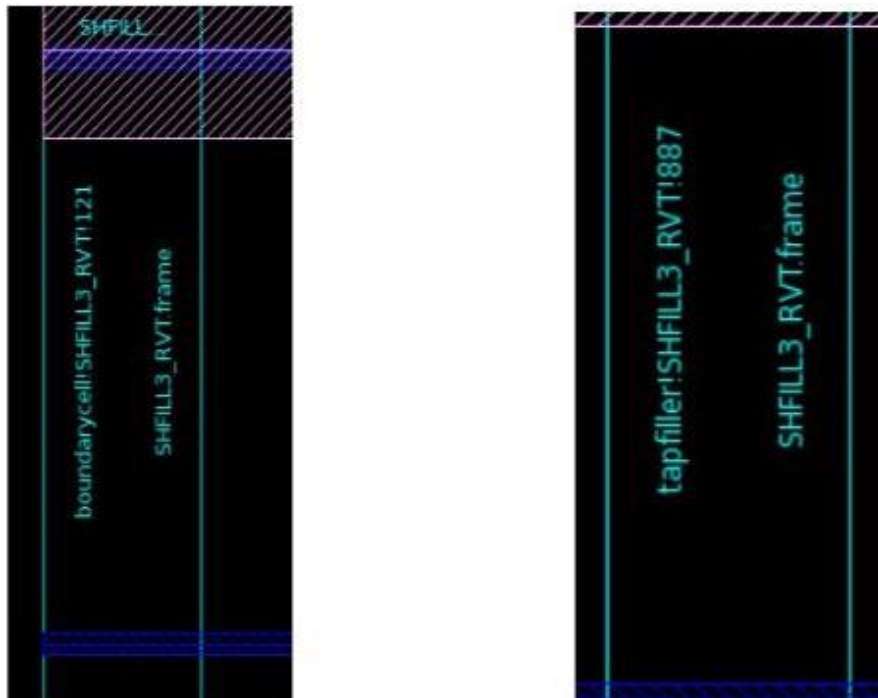


Fig 1: End cap cell and Tap cell

Check legality for End Cap and Tap Cells

Check the placement of End cap cells and tap cells are proper and no violations, run the following commands,

Study the reports and check any tap cells and end cap cells placement related violations.

```

*****
Report : Legality
*****

VIOLATIONS BY CATEGORY:
MOVABLE  APP-FIXED  USER-FIXED  DESCRIPTION
5         0         0         Two objects overlap.
0         0         0         A cell violates a pnet.
0         0         0         A cell is illegal at a site.
0         0         0         A cell is not aligned with a site.
0         0         0         A cell has an illegal orientation.
0         0         0         A cell spacing rule is violated.
0         0         0         A layer rule is violated.
0         0         0         A cell is in the wrong region.
0         0         0         Two cells violate cts margins.
0         0         0         Two cells violate coloring.
5         0         0         TOTAL

TOTAL 5 Violations.

VIOLATIONS BY SUBCATEGORY:
MOVABLE  APP-FIXED  USER-FIXED  DESCRIPTION
5         0         0         Two objects overlap.
5         0         0         Two cells overlap.
0         0         0         Two cells have overlapping keepout margins.
0         0         0         A cell overlaps a blockage.
0         0         0         A cell keepout margin overlaps a blockage.
0         0         0         A cell violates a pnet.
0         0         0         A cell is illegal at a site.
0         0         0         A cell violates pin-track alignment rules.
0         0         0         A cell is illegal at a site.
0         0         0         A cell violates legal index rule.
0         0         0         A cell has the wrong variant for its location.
0         0         0         A cell is not aligned with a site.
0         0         0         A cell is not aligned with the base site.

```

Saving the Block:

Save the changes done to the block as `floorplan_done`

```
icc2_shell> save_block -as floorplan_done
```

Close the block and Library before exiting from the `icc2_shell`.

```
icc2_shell> close_blocks -force
```

```
icc2_shell> close_lib
```

```
icc2_shell> exit
```